

DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY
PRACTICE SHEET 3

Prerequisite	Traversal, comparison, counting, sequence detection, index tracking, boundary tracking, difference calculation, pattern recognition and maximum/longest sequence identification.
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1. Server Response Time – Performance Degradation Detector

A cloud monitoring system records response times for N consecutive requests. Identify the number of times the response time increases continuously for three consecutive requests. Also find the largest increase between two consecutive requests.

Input Format

N

response_time[0] response_time[1] ... response_time[N-1]

Sample Input

7

120 140 160 130 180 200 250

Processing

- 120 < 140 < 160 → Performance degradation pattern
- 130 < 180 < 200 → Performance degradation pattern
- 180 < 200 < 250 → Performance degradation pattern

Largest consecutive increase:

130 → 180 = 50 ms

Sample Output

Degradation Patterns: 3

Largest Increase: 50 ms

2. Data Center CPU Usage – Overload Sequence Detector

A data center stores CPU utilization for N consecutive time intervals. A server is considered to have an overload sequence if CPU utilization is greater than 80% for three or more consecutive intervals. Find:

- Number of overload sequences
 - Longest overload sequence
-

Sample Input

10

60 85 90 95 70 82 88 65 91 93

Processing

Sequences above 80:

85 90 95 → Length 3

82 88 → Length 2

91 93 → Length 2

Sample Output

Overload Sequences: 1

Longest Overload Duration: 3 intervals

3. Network Traffic – Traffic Spike Detector

A network system records traffic for N intervals. A traffic spike occurs when the current traffic is at least 50% greater than the previous interval. Find:

- Number of spikes
 - Largest spike
 - Position of the largest spike
-

Sample Input

6

100 180 200 90 150 300

Processing

100 → 180 = Spike

90 → 150 = Spike

150 → 300 = Spike

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Sample Output

Number of Spikes: 3

Largest Spike: 150 MB

Position: 6

4. E-Commerce Sales – Best Growth Period

An e-commerce company stores daily sales for N days. Find the longest consecutive period where sales continuously increase. Also calculate the total growth during that period.

Sample Input

8

1000 1200 1500 1300 1400 1600 1800 1700

Processing

Increasing sequences:

1000 → 1200 → 1500

Length = 3

1300 → 1400 → 1600 → 1800

Length = 4

Total growth:

1800 - 1300 = 500

5. API Failure Monitor – Failure Recovery Analyzer

An API monitoring system records failed requests every hour. A failure recovery occurs when the failure count decreases for three consecutive hours. Find the longest recovery sequence.

Sample Input

8

100 80 60 40 70 50 30 20

Processing

100 → 80 → 60 → 40

Recovery length = 4

70 → 50 → 30 → 20

Recovery length = 4

Sample Output

Longest Recovery Sequence: 4 hours

6. Employee Productivity – Performance Ranking with Duplicate Scores

An IT company stores productivity scores of N employees. Find:

- The highest score
 - The second-highest distinct score
 - Number of employees who achieved the highest score
 - Rank of each employee based on distinct scores
-

Sample Input

6

78 92 85 92 88 75

Sample Output

Highest Score: 92

Second Highest Score: 88

Employees with Highest Score: 2

Ranks:

Employee 1: Rank 5

Employee 2: Rank 1

Employee 3: Rank 4

Employee 4: Rank 1

Employee 5: Rank 2

Employee 6: Rank 6

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7. Cybersecurity – Suspicious Login Pattern Analyzer

A security system stores failed login attempts for N consecutive time periods. A security alert occurs if: Failed attempts are greater than or equal to 5 for three consecutive periods. Find:

- Number of alerts
- Longest suspicious sequence
- First position where an alert occurs

Sample Input

10

2 6 7 8 3 5 9 10 4 6

Processing

6 7 8 → Alert

5 9 10 → Alert

Sample Output

Security Alerts: 2

Longest Suspicious Sequence: 3

First Alert Position: 2

8. Cloud Storage – Usage Anomaly Detector

A cloud platform stores daily storage usage for N days. A day's usage is considered an anomaly if it differs from the average usage of its immediate previous and next days by more than 50 GB. Ignore the first and last elements.

Sample Input

7

100 110 300 115 120 125 130

Processing

For 300:

Average of neighbours = $(110 + 115) / 2 = 112.5$

Difference = $300 - 112.5$

Since the difference is greater than 50, it is an anomaly.

Sample Output

Anomalies Detected: 1

Anomaly Position: 3

9. Software Testing – Bug Trend Reversal Detector

A software testing system stores the number of bugs found during N testing cycles.

Detect a trend reversal when:

- Bug count was continuously increasing and then starts decreasing, OR
- Bug count was continuously decreasing and then starts increasing.

Sample Input

8

10 15 20 25 18 12 15 20

Processing

10 → 15 → 20 → 25 Increasing

25 → 18 → 12 Decreasing

Trend reversal detected at:

25

Then:

12 → 15 → 20

Another reversal occurs.

Sample Output

Trend Reversals: 2

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10. DevOps Deployment – Performance Stability Analyzer ★★★★★

A DevOps team records deployment times for N consecutive software releases. A deployment is considered unstable if its deployment time differs by more than 30% from the previous deployment.

The system must:

- Count unstable deployments.
- Find the longest stable sequence.
- Find the largest percentage change.

Sample Input

8

100 110 105 200 190 195 300 290

Processing

100 → 110 = 10% change → Stable

110 → 105 = 4.5% change → Stable

105 → 200 = 90.5% change → Unstable

Continue this processing for the entire array.

Sample Output

Unstable Deployments: 2

Longest Stable Sequence: 3

Largest Percentage Change: 90.48%

Skills Acquired:	Array traversal, comparison, searching, sorting, sequence detection, index tracking and pattern recognition.
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